Drilling

Drilling holes deep below Earth's surface makes it possible to access natural resources such as water, oil, and gas. Holes are also drilled for scientific purposes, for example to remove ice core samples that can be analyzed to provide information about past environmental conditions.

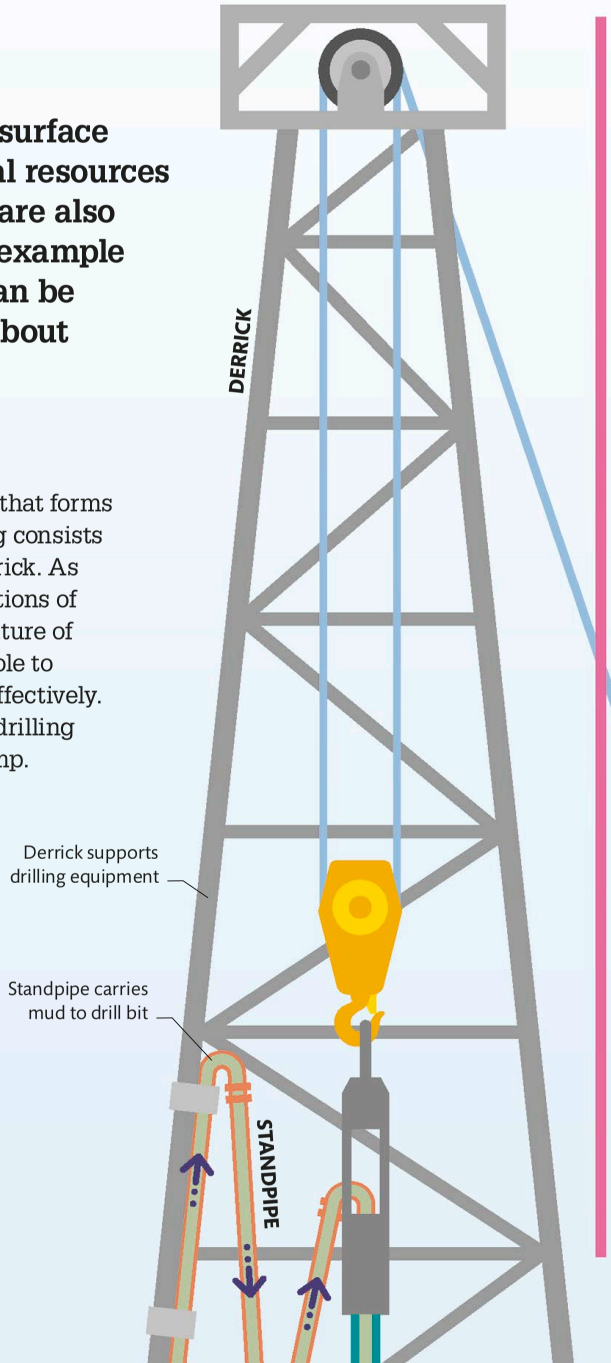
Drilling for oil

Oil is a naturally occurring organic substance that forms liquid deposits underground. An oil drilling rig consists of a drill supported by a structure called a derrick. As the drill moves down through the ground, sections of steel casing are placed around the hole. A mixture of fluid known as mud is also pumped into the hole to make the cutting tool, or drill bit, work more effectively. Once the drill reaches the oil, the derrick and drilling equipment are removed and replaced by a pump.

DRILLING ICE CORES

Ice forms from the gradual buildup of snow, so lower layers are older than upper ones, and analyzing ice cores can provide information about past climatic conditions. Ice cores are drilled with a hollow pipe, and some can be 2 miles (3 km) deep.





Offshore drilling

To access oil deposits below the seabed, oil companies use specialized mobile offshore drilling units (MODUs). Once an oil deposit has been found, some MODUs can be converted into oil production rigs. Usually, however, a MODU is replaced with a more permanent oil production platform after oil has been struck.



Jackup
A jackup is a MODU with legs that can be extended down to rest on the seabed. This keeps the rig safe from tidal motions and waves.



Semisubmersible
Semisubmersibles float on the sea surface on top of submerged pontoons. Some can convert to production rigs once oil is found.



Drillship
These are specialist ships with a drillship rig on the top deck. The drill operates through a hole in the hull. Drillships can operate in deep water.



Drilling barge
A drilling barge is a small vessel fitted with a rig raised off its deck. Drilling barges are suitable for use only in calm, shallow water.